Data Customization-Based Multiobjective Optimization Pruning Framework for Remote Sensing Scene Classification

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***Abstract*— Pruning techniques have been utilized widely for convolutional neural networks (CNNs) to reduce the computation resources in remote sensing scene image classification. However, conventional pruning techniques are weight-based, which cannot balance the pruning ratio and representation ability appropriately. In this article, we propose a data customization-based multiobjective optimization pruning (DCMOP) framework for the pruning in remote sensing scene image classification, which can not only tradeoff between pruning ratio and capability for CNNs but also speed up the evolutionary process for the pruning. We adopt the multiobjective evolutionary algorithms (MOEAs) to search for a tradeoff between the pruning ratio and capability for CNNs. However, a big concern of pruning for networks via MOEAs is that the evaluation of subnetworks is time-consuming. This originates that the slimmed subnetworks require a lot of retraining operation, which will burden the hardware. In order to alleviate this limitation, we design a data customization-based proxy mechanism (DCPM) to reduce the size of the input dataset in terms of the structure of the slimmed subnetwork to accelerate significantly the evolutionary process for the pruning. According to this, our proposed DCMOP achieves the pruning with higher efficiency and performance by cooperating with MOEAs and DCPM. Experimental results based on four datasets of AID, NWPURESISC45, PatternNet, and WHU-RS19 show that the proposed DCMOP can achieve a balance between model performance and pruning rate, while obviously reducing the time cost of the pruning.**

***Index Terms*— Data customization, filter pruning, multiobjective optimization, remote sensing scene classification.**

I. Introduction

With the development of deep techniques, convolutional neural networks (CNNs) [1] have been widely used in remote sensing images scene classification, such as environmental planning [2], [3], resource exploration [4], [5], geological hazard assessment [6], [7], and land cover survey [8], [9], which has generated impressive application value.

Due to the real-time constraint for the remote sensing scene classification on the satellite, computing resources are highly required to support the application of CNNs, which may burden the edge devices obviously. In particular, CNNs with complicated structures will generate a huge of FLOPs and parameters when remote sensing scene images are processed, i.e., there are 2.06G FLOPs when dataset AID [10], NWPU-RESISC45 [11], PatternNet [12], and WHU-RS19 [13] are tackled by ResNet-50 [14].

Therefore, the compression of models is particularly necessary. Currently, pruning methods and neural architecture search (NAS) techniques have been proven to be very effective in reducing the model size. Pruning methods aim to remove redundant parameters or filters from the original model, thereby reducing the parameters and FLOPs correspondingly.

The NAS can automatically generate an optimal network by combining filters with a given search space [15], [16], [17], which can customize a compact model in terms of the specified task. However, the search process is tough because the evaluation of subnetworks is time-consuming, and therefore, pruning techniques are utilized to slim models on the satellite in this article.

However, traditional pruning methods only focus on filter reduction or accuracy recovery, which makes it difficult to strike a balance between the accuracy and the pruning rate of the model. Currently, most of the traditional pruning methods that are used to save computing resources on satellite platforms can be divided into two categories: structural pruning and unstructured pruning [18], [19], [20]. Unstructured pruning methods focus on forcibly converting convolutional weights and feature maps to sparse matrices, which does not completely delete the whole filter, and then cannot reduce the size of feature maps. Although structural pruning methods can reduce the number of parameters and the size of FLOPs, they are usually sensitive to weight, which means that the pruning rate is generally a manually designed hyperparameter. It implies that it is hard to confirm an appropriate pruning rate to prune the networks maximally, as well as maintain higher accuracy.

In order to effectively balance the contradiction between the accuracy and pruning rate of the model, we adopt MOEAs to build a pruning framework to improve the performance of

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the slimmed model on the satellite. Because MOEAs [21], [22] usually can balance multiple conflicting goals to find the optimal solution, and therefore, multiobjective optimization pruning framework (MOPF) is effective for balancing the accuracy and pruning rate of the model. Nevertheless, the evaluation of subnetworks during the evolutionary process is time-consuming due to the retraining of these sparse individuals, and then burdens the hardware, especially in the satellite platform.

In order to solve this dilemma, we design an adaptive data customization mechanism to speed up the evaluation of subnetworks during the evolutionary pruning process. Data customization means we will feed a specified quantity of data to the network according to the size of the searched subnetwork, and it can save computation resources by reducing the retraining of each individual.

The data customization is inspired by the work from Bubeck and Sellke [23], which derived smoothing by computing the required parameters to fit data points to curves with equivalent robustness. In general, parametric model classes can be used for data interpolation as long as the number of parameters is greater than the number of equations to be satisfied. This also means that the size of the network determines the amount of data it requires. Since the size of the network is different in terms of distinct pruning degrees, the amount of data required for different pruning degrees of the network is also distinguished. Inspired by the above conclusion, it is possible to customize the datasets for different sparse individuals to accelerate the evaluation of subnetworks. Furthermore, we found that over-parameterization not only improves the efficiency of the model training process but also improves the generalization ability of the network.

In this article, we propose a data customization-based proxy mechanism (DCPM) to accelerate the evaluation of subnetworks during the evolutionary pruning process.

Initially, we generate subnetwork structures in terms of initialization of the original model, as well as code the structure of these subnetworks by genotypes with "0–1." Then, the data customization mechanism is utilized to retrain subnetworks during the evolutionary process, which can speed up the evaluation of individuals impressively.

In addition, we capture the quality of pruned subnetworks by incomplete training, which in turn can further speed up the evaluation of subnetworks. Namely, an incompletely trained subnetwork is regarded as the proxy model to reflect the quality of the full subnetwork. This article resorts AID [10], NWPU-RESISC45 [11], PatternNet [12], and WHU-RS19 [13] classic datasets to VGG-16 [24] and ResNet-50 [14] models to find that the pruning rate of the model is maximized on different datasets with guaranteed accuracy, which can show the feasibility and superiority of this framework.

Three contributions are summarized in this article as follows.

1) We adopt a MOPF to balance the pruning rate and accuracy of the model and obtain the redundancy of filters to achieve global optimization of the model. This framework not only reduces a massive of computation resources but also maintains higher accuracy for the remote seining scene image classification.
2) We design a DCPM to accelerate the evaluation of subnetworks during the multiobjective evolutionary pruning process. The efficacy of DCPM is proven by extensive experiments and sufficient mathematical analysis.
3) Data customization-based multiobjective optimization pruning (DCMOP) yields impressive pruning results on VGG-16 [24] and ResNet-50 [14], as well as balance the accuracy and pruning rate of the model well on four remote sensing datasets: AID [10], NWPU-RESISC45 [11], PatternNet [12], and WHU-RS19 [13].

This article is written as follows. In Section II, we discussed the related work of MOEAs and other remote sensing classification technologies, as well as the analysis of traditional pruning methods and NAS-based methods. In Section III, we introduced and explained our pruning framework (DCMOP). In Section IV, we verify our method on different datasets and models and make a visual analysis. Finally, we summarize the framework and the whole work.

II. Related Work

In this section, we will divide the content into three parts. First, we will discuss the relevant technologies and analysis of image scene classification in the field of remote sensing. Second, we will summarize and analyze the existing model compression technologies, mainly including the pruning method and NAS-based techniques. Finally, we will give a detailed introduction and explanation of the motivation for using multiobjective optimization in this article.

A. Relevant Technologies of Remote Sensing Image Scene Classification

With the rapid increase in the number of remote sensing images and the continuous improvement of their complexity and resolution, the early methods that only rely on manual feature extraction gradually fall behind, and the related technologies [25], [26], [27], [28], [29] based on depth learning and depth learning also gradually mature and have achieved good results in the field of image processing.

ResNet [14], AlexNet [30], VGGNet [31], and other deep learning frameworks are more widely used in the research of deep scene classification. Cheng et al. [32] used three different CNNs for classification, which is to use the pretrained CNN [32] as a feature extractor, and then optimize the model through the dataset to classify. Shawky et al. [33] proposed a method based on CNN-MLP to classify remote sensing images. The author extracted features from CNN [32] without full connection, and classified images on a multilayer perceptron relying on the Adagrad optimizer, which achieved good results. Qu et al. [34] used the classification function of support vector machines (SVMs) to classify the features extracted by SVM and then learn the classification function to achieve classification. Zhou et al. [35] proposed the ResNet-TP method for feature extraction and classification.

Combined with the model used in the above methods, it can be concluded that the complex model structure cannot be

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flexibly applied to the specified dataset, and the hardware requirements are very high. Therefore, the compression of the network is particularly important.

B. Network Compression Techniques

Currently, pruning technology and NAS-based methods are commonly used among the existing network compression technologies.

1) Traditional Pruning Technology: Pruning technology is a more compact model representation by deleting weight parameters or complete filters. The traditional pruning method is divided into structured pruning and unstructured pruning. For unstructured pruning, Han et al. [36] set a threshold to distinguish the importance of weights to calculate their redundancy and prune them. In the pruning process, the network repeats fine-tuning and training until its accuracy is reduced. Han et al. [37] proposed to make improvements in pruning, quantization, and Huffman coding, and proposes a strategy of combining pruning, quantization, and Huffman coding to prune, realizing network compression without losing accuracy. For structural pruning, Liu et al. [38] achieved the purpose of model compression by pruning the number of channels in each layer. Huang and Wang [39] introduced additional scaling factors to adaptively train the output of different dimensions for tailoring.

2) NAS-Based Technology: NAS is a new method of automatic learning network structure, which is used to reduce the heavy network design cost. Zheng et al. [40] designed a framework of AutoCL to find a better structure by using the design hierarchy search space. Peng et al. [41] proposed a greedy progressive strategy, which introduces bi-hierarchical optimization and designs a continuous search space to find the optimal structure. Zoph and Le [15] trained neural networks using reinforcement learning and searches for the optimal structure on special datasets. Real et al. [16] combined evolutionary algorithm (EA) with NAS, which can well solve the design problem of neural network structure, and cannot be lower than the manually designed structure in image classification. Liu et al. [17] mixed candidate operations using the softmax function to transform the search space into a continuous space, and then used gradient-based optimization methods to find the optimal structure. However, although the NAS-based method can find better structures, its relatively large search space leads to high search costs and time costs, which in turn wastes the computing resources of edge devices.

C. Multiobjective Optimization EAs

It is well known that the Pareto optimal solution of the multiobjective optimization problem (MOP) [42], [43] can be transformed into an optimal solution of an objective optimization problem that aggregates all objective functions under certain conditions.

MOP can be stated as follows:

$$\begin{aligned} &\min \quad F(x) = (f_1(x), f_2(x), \ldots, f_m(x))^{\mathrm{T}}, \\ &s.t. \quad x \in \Omega \subseteq \mathbb{R}^n \end{aligned} \tag{1}$$

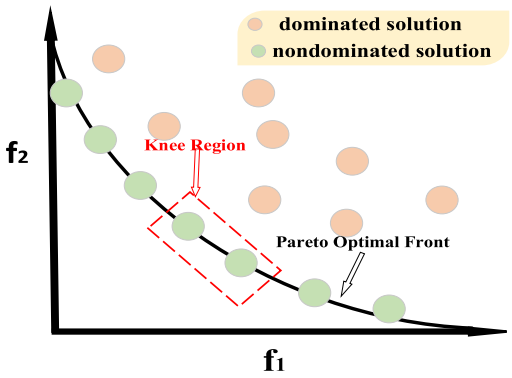


Fig. 1. Schematic of Pareto front. Among them, the orange circle represents the dominant solution x, the green circle represents the nondominant solution x^*, different x^* forms the Pareto front, and the red dotted line represents the knee region.

where Ω represents the decision space, m is the scale of the objective function, and $x = (x_1, x_2, \ldots x_n)^{\mathrm{T}}$ is a vector of the n-dimensional decision space. $f(x)$ is the objective function of MOPs. $F(x)$ is the mapping function that maps the decision space Ω to the m-dimensional objective space Θ. That is, $F(x) : \Omega \to \Theta \subseteq R^m$.

Generally, MOP is defined as the minimization of multiple objectives. The dominance of the decision variables x_u and x_v is as follows:

$$\begin{cases} \forall i = 1, 2, \ldots n, \ f_i(x_u) \le f_j(x_v), \\ \wedge \exists j = 1, 2, \ldots, n, \ f_j(x_u) \le f_j(x_v) \end{cases} \tag{2}$$

where the decision vector x_u is said to dominate another vector x_v, and $x_u, x_v \in \Omega$. If there is no other solution x in (2) to dominate x^*, then x^* is called Pareto optimal solution, where $x^* \in \Omega$. F is used as the Pareto optimal target vector. The set of Pareto optimal objective vectors is called Pareto optimal front (PF) [44].

Because EAs can well balance MOPs, many multiobjective EAs (MOEAs) have been proposed to solve MOPs by finding a set of Pareto optimal solutions [21] through different methods. Eberhart et al. [45] and [46] presented the particle swarm optimization (PSO) to solve MOPs, which is most classic for its simple convergence speed and algorithm complexity. The Pareto front vectors and objective vectors are listed in Fig. 1, where the objective vector of the front is nondominant. The area bounded by the red dotted line in the figure is called the knee region. Usually, in this area, we can obtain the best plan to balance the conflicting goals, that is, to improve the situation that the progress of one goal leads to the degradation of another goal.

In the traditional pruning method of the network, a truncation parameter for pruning is usually introduced to balance the accuracy of remote sensing scene classification and the pruning rate of the network. This method requires continuous manual adjustment of parameters for training, so as to obtain an appropriate solution, which is very difficult and lacks robustness. Specifically, when the remote sensing scene dataset containing heavy noise is classified, these noises often affect the training, and then cause important filters to be deleted.

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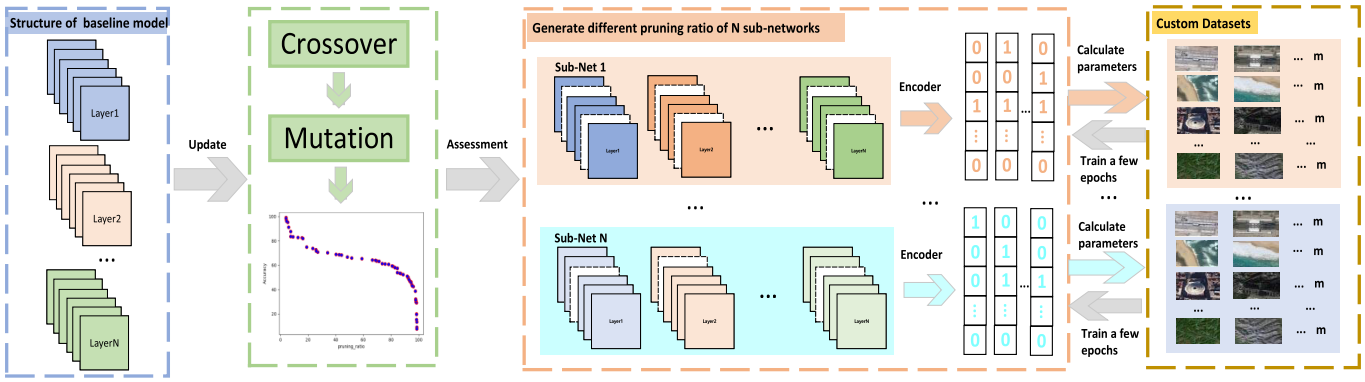


Fig. 2. Overview of DCMOP. (Left) DCMOP is proposed to prune the networks, including the initialization of the network, and generate new subnetworks through crossover, and mutation with the "0–1" coding strategy. (Right) We design an adaptive data customization mechanism to speed up the evaluation of subnetworks during each evolutionary process.

With the increase of truncation parameters for pruning, the pruning network usually performs poorly.

In this article, we propose a MOPF to solve this problem. MOPF solves the pruning problem by using multiobjective optimization. We balance the accuracy and pruning rate of the model by optimizing the loss and pruning rate of each layer of filters. In training, the importance of feature learning of the filters determines whether it is redundant. The pruning ratio for pruning usually controls the proportion of all filter deletions in the network. That is, the higher the pruning ratio is, the higher the redundancy of the filter in the model is, and the more filters are pruned. More recently, because of the need to train models with different pruning rates for iteration, we can find the pruning model with the best accuracy. Therefore, MOPF implicitly represents the weight vector of the decomposed MOP by generating subnetworks with different pruning rates. MOPF expands the search scope of pruning rate model selection, making the evaluation process of the pruning model more global. Due to the defect that traditional pruning methods need to manually select pruning parameters or extract prior knowledge. In order to effectively solve this problem, MOPF can not only automatically generate PF, but also effectively achieve the balance between model accuracy and pruning rate.

III. Methodology

In this article, we adopt a DCMOP framework to balance the precision and pruning rate of the model for the network pruning. The overall framework is shown in Fig. 2.

In the left part, a multiobjective pruning framework is proposed to prune the networks, including the initialization, crossover, and mutation with the "0–1" coding strategy. In the right part, we design an adaptive data customization mechanism to speed up the evaluation of subnetworks during each evolutionary process. The detail of this framework is illustrated in the following parts separately.

A. Multiobjective Optimization Pruning Framework

Pruning of the model usually refers to removing redundant filters in the convolution layer and leaving filters with higher importance. It means if the pruning rate of the filter is over-pruned, some filters that contribute significantly to the overall performance of the network will be pruned, and then the learning ability of the whole model will be degraded. Notably, the precision and the pruning rate are contradictory, and it is difficult to determine an appropriate learning rate for the previous weight-norm-based methods. Therefore, the MOPs are utilized to yield a balanced tradeoff between the pruning rate and accuracy.

In this article, we introduce multiobjective optimization into the pruning model, thus establishing a MOPF. Impressively, MOPF can optimize the pruning rate of the model while ensuring the accuracy of the model, i.e., MOPF can yield a set of balanced pruning rates and accuracy along the PF, and then provide us with potential candidates to pick up the most appropriate slimmed model. The objective function of pruning rate M_{pr} and accuracy M_{acc} of the model can be written as follows:

$$\begin{cases} \min F(p) = \{M_{\text{pr}}(p_i), M_{\text{acc}}(p_i)\}, \\ s.t. \quad p \in \Omega, i = 1, 2 \ldots, N \end{cases} \tag{3}$$

where p represents the subnetwork and i represents the N-dimensional decision space. $F(x)$ represents the mapping function from decision space Ω to 2-D target space Θ.

In particular, there are several important steps in MOPF as follows.

1) Initialize the subnetwork and then update them through crossover and mutation.

In the beginning, initialization generates subnetworks with different pruning degrees based on the initial model. The structure of each subnetwork can be represented with "0–1" encoding. Specifically, "0" and "1" mean the corresponding filters will be removed and remain, respectively. The structure of the subnetwork can be written as follows:

$$f = O^L([\text{rand}_{(0,1)}(\mathbf{f}_1)], \ldots, [\text{rand}_{(0,1)}(\mathbf{f}_L)]) \tag{4}$$

where $O^L()$ represents the set of filters across the L-dimensional convolution layer. $\text{rand}_{(0,1)}(\mathbf{f}_i)$ represents random coding the ith convolutional layer from 0 to 1, where $\mathbf{f}_i$ contains the filter vectors of all filters in the ith layer.

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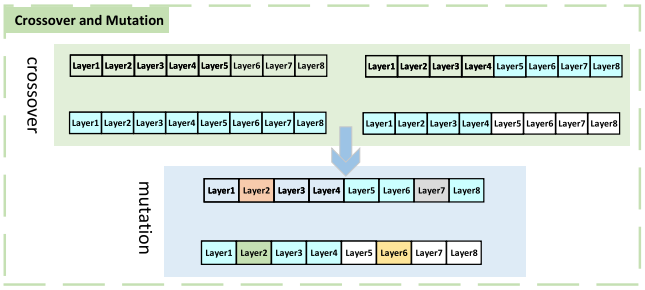


Fig. 3. Details of crossover and mutation. In the first part, the crossover stands for the filter exchange between two subnetworks across all layers. In the second part, the mutation operation usually reverses some filters to improve the diversity through local optimization.

Then, genetic operators crossover and mutation are used to update the next generation during evolution. The specific details of crossover and mutation are shown in Fig. 3. In this part, the crossover stands for the filter exchange between two subnetworks across all layers, i.e., the exchange emerges between the i-layer of subnetwork p_u and the j-layer of subnetwork p_v to enrich the population diversity.

Moreover, the mutation operation usually reverses some filters to improve the diversity through local optimization, i.e., some filters are mutated to be removed to generate new architecture f_{new} to measure the quality, and it can be written as

$$f_{\text{new}} = \begin{cases} f^u_{\text{new}}(M^{\text{pr}}(p^{\text{new}}_u), M^{\text{acc}}(p^{\text{new}}_u)), \\ f^v_{\text{new}}(M^{\text{pr}}(p^{\text{new}}_v), M^{\text{acc}}(p^{\text{new}}_v)). \end{cases} \tag{5}$$

2) Adopt the "0–1" encoding to convert subnetwork structures with different pruning degrees into digital genotypes.

Subnetworks are generated by crossover and mutation via "0–1" coding, which can transfer a network from phenotype to genotype. Then, we denote the evolutionary process as follows. In the first part, the process of crossover can be described as follows:

$$\begin{cases} p_u = [\mathbf{f}_1([0, 1, 1, \ldots, 0, 1]), \ldots, \mathbf{f}_K([1, 0, 1, \ldots, 0, 0])], \\ p_v = (\mathbf{f}_1([0, 0, 1, \ldots, 0, 1]), \ldots, \mathbf{f}_K([0, 0, 1 \ldots, 0, 0])] \end{cases} \tag{6}$$

where p_u and p_v represent the uth and vth subnetworks, respectively. "0" and "1" means the corresponding filters will be removed and remain, respectively. Then, p^{new}_u and p^{new}_v can be written as: crossover can be described as follows:

$$\begin{cases} p^{\text{new}}_u = [\mathbf{f}_1([1, \boxed{\mathbf{0}}, 1, \ldots, 0, 1]), \ldots, \mathbf{f}_K([0, \boxed{\mathbf{1, 0}}, \ldots, 0, 0])], \\ p^{\text{new}}_v = [\mathbf{f}_1([0, \boxed{\mathbf{1}}, 1, \ldots, 0, 1]), \ldots, \mathbf{f}_K([1, \boxed{\mathbf{0, 1}} \ldots, 0, 0])]. \end{cases} \tag{7}$$

In the second part, the mutation operation can be described as

$$p^u_{\text{new}'} = [\mathbf{f}_1([1, \boxed{\mathbf{1}}, 1, \ldots, 0, 1]), \ldots, \mathbf{f}_K([1, \boxed{\mathbf{1, 1}} \ldots, \mathbf{0}, 0])]. \tag{8}$$

3) Evaluate all individuals and select the most suitable one from the converged PF.

After the crossover and mutation operation, all subnetworks are evaluated to select the individual that best balances the pruning rate of the model and the accuracy of the model. During the evaluation process, we design a DCPM to speed up the evaluation of subnetworks.

B. Data Customization-Based Proxy Mechanism

In this section, we divide DCPM into two parts for a detailed introduction. In the first part, the data customization mechanism is introduced to accelerate the evaluation speed. The second part describes mainly the agent model mechanism.

1) Data Customization for Improving Training Efficiency: It is hard to evaluate each subnetwork by full training because of limited computation resources. Inspired by the conclusion that the data size determines the number of parameters from [16], [47], and [48], we can customize the input data in terms of the size of subnetworks to reduce the consumption of evaluation, and then improve the efficacy of MOPF.

Considering the map between the dataset and model size, the dimension D of the dataset that is sent can be quantified by the number of parameters for the subnetwork

$$\text{param}_i = \text{data}_i \times D \tag{9}$$

where param_i is the number of parameters for subnetwork i, and data_i represents the size of the dataset that will be fed into ith subnetwork.

Correspondingly, the new input dataset $\mathcal{D}_{p,q}$ for the pth subnetwork in qth evolutionary iteration can be customized as

$$\begin{aligned} &\mathcal{D}_{p,q} = \mathcal{D}_{[0,1,..,B]}, \\ &s.t. \quad B = \lfloor \frac{\text{param}_{p,q}}{\mathcal{H}\mathcal{W}} \rfloor \end{aligned} \tag{10}$$

where the subscript $[0, 1, 2, \ldots, B]$ means the corresponding images will be used to construct the new dataset, and $\lfloor \cdot \rfloor$ will calculate the upper integer of the specified number. $\mathcal{H}$ and $\mathcal{W}$ stands for the height and width, respectively. The $\text{param}_{p,q}$ stands for the number of the parameters for pth subnetwork in qth iteration. Consequently, the retraining epochs of the subnetwork can be reduced apparently due to fewer data, which can evaluate each subnetwork with plentiful efficiency without the assistance of massive computation resources.

Furthermore, the dataset of remote sensing scene classification contains kinds of categories, and feature information may be lost if some categories are not fed sufficiently.

In order to fully solve this problem, we will take the amount of incoming data according to each type of dataset as shown in the right half of Fig. 2. The calculation method can be written as follows:

$$B^{\text{per}} = \frac{B}{A} \tag{11}$$

which B^{per} stands for each type of dataset of subnetwork i and A represents the total category of this dataset.

2) Proxy Mechanism: Previous works have concluded that there is a positive correlation between fully trained and incompletely trained subnetwork models [16], [47], [48], this means that incompletely trained subnetwork models can reflect the quality of fully trained subnetwork models. Therefore,

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we accelerate the evaluation process of the pruned subnetwork by data customization over incomplete training. Specifically, we use the obtained under-trained subnetworks as proxy models to replace the subnetworks that need to be fully trained.

The loss of incomplete-trained and fully-trained pruned models can be denoted $L^{\text{under}}(\text{data}_i, f_l^i = 0)$ and $L^{\text{under}}(\textbf{data}, f_l^i = 0)$, respectively. According to [18], the loss between the original and pruned model can be described as

$$\begin{cases} \Delta L = |L(\textbf{data}, f_l^j) - L(\text{data}_i, f_l^j = 0)|, \\ s.t. \quad l = 1, 2, 3, \ldots, L \end{cases} \tag{12}$$

where data_i and **data** represent the input dataset for the subnetwork i and original model, respectively. f_l^j stands for the jth filter in layer l will be removed.

By decomposing (12) through the Taylor expansion at the first-order, we can rewrite the equation as

$$\begin{aligned} \Delta L &= |\frac{\partial L}{\partial h_l} h_l|, \\ &= |\frac{1}{\text{param}_i} \sum_n \frac{\partial L}{\partial h_{l,n}^j} h_{l,n}^j|, \\ &= |\frac{1}{\text{param}_i} \sum_n (\sum_k \frac{\partial L}{\partial h_{l,n}^k} h_{l,n}^k) - \sum_{k^*} \frac{\partial L}{\partial h_{l,n}^{k^*}} h_{l,n}^{k^*})|, \end{aligned}$$

$$\begin{aligned} s.t. \sum_{k^*} h_{l,n}^{k^*} &= h_{l-1}^{j',n} \bigotimes f_l^{k^*}, \\ &= \sum_{k^*} \sum_n \sum_{j'} h_{l-1}^{j',n} f_l^{k^*}, \quad i \neq k^* \end{aligned} \tag{13}$$

where param_i is the number of parameters for a feature map in subnetwork i, $h_{l,n}^j$ represents the nth parameter of jth output feature in layer l. $\bigotimes$ stands for the convolutional operation. $\sum_k$ and $\sum_{k^*}$ stand for the summation of all the filters and remaining filters after pruning. $h_{l-i}^{j',n}$ represents the nth region in j'th output feature for the layer $l-1$, and $f_l^{k^*}$ is k^*th filter in layer l. Equation (13) can be regarded as the product of the weight based on the first-order Taylor expansion and the output characteristic gradient.

Therefore, combining (12) and (13), we can get the proxy model $(L(\text{data}_i, f_l^j = 0)^{\text{proxy}})$ and fully-trained model $(L(\textbf{data}, f_l^j = 0)^{\text{fully}})$ as follows:

$$L(\text{data}_i, f_l^j = 0)^{\text{proxy}} = L(\text{data}_i, f_l^j) \pm |\frac{\partial L}{\partial h_l} h_l|, \tag{14}$$

$$\begin{aligned} L(\textbf{data}, f_l^j = 0)^{\text{fully}} &= \frac{\partial L(\text{data}_i, f_l^j = 0)^{\text{proxy}}}{\partial f_l^*}, \\ &= \frac{\partial L(\text{data}_i, f_l^j) \pm \partial |\frac{\partial L}{h_l} h_l|}{\partial f_l^*} \end{aligned} \tag{15}$$

where f_l^* represents the set of remaining filters after pruning for layer l, and rewrite $L(\text{data}_i, f_l^j = 0)^{\text{proxy}}$ as $L(\text{data}_i, f_l^*)$.

Algorithm 1 DCMOP

Input: MOP: Pruning ratio: M^{pr}; Accuracy: M^{acc}.
Iterations: $G = 50$;
Population(the size of sub-networks): $N = 50$;
Sub-network: p_i, $i = 1, \ldots, N$;
The number of neighbors for each weight vector: $T = 10$;
Weight vector: $\lambda = \{\lambda_1, \ldots, \lambda_N\}$.
Output: ***EP***
Step 1: Initialize:
EP=0;
Random generation of sub-networks with different pruning rates according to Equation (4): p_i, $i = 1, \ldots, N$;
Generate N weight vectors: $\lambda_1, \ldots, \lambda_N$;
Calculate the Euclidean distance between each vector and select T vectors as neighbors, such as: $f(p_1) : \{f(p_{T_1}), \ldots, f(p_{T_T})\}$;
Calculate the parameter quantity of each sub-network: $param$;
Initialize $z = \{z_{pr}, z_{acc}\}$ to save the optimal solution.
Step 2: Evolution:
for $t < G$ **do** *Evolutionary loop*
  for $i = 1, \ldots, N$ **do** *Evolutionary loop*
    <1>: Calculate the T neighbors of the $i-th$ individual: $f(p_{T_1}), \ldots, f(p_{T_T})$;
    <2>: Select two individuals p_u and p_v from p_T;
    <3>: Crossover and mutate of p_u and p_v to generate new individuals f_{new} according Equation (5) ;
    <4>: **Encoding model according to Equation (4) and calculate the amount of data ($data_i$) required by the model according to Equation (9) and generate the proxy model;**
    <5>: Train a few epochs and generate a new surrogate model according to the agent mechanism: $f_{new'}^u(M^{pr}(p_{new'}^u), (M^{acc}(p_{new'}^u))$ and $f_{new'}^v(M^{pr}(p_{new'}^v), (M^{acc}(p_{new'}^v))$;
    <6>: Through comparing between $f_{new'}^u$, $f_{new'}^v$ and f^u, f^v, the non-dominated individuals p^* is selected as the parent of the next generation to generate a new individual;
    <7>: Update z;
    <8>: Update EP with the non-dominated individuals;
    $i++$.
  end for
  $t++$.
end for

C. Implementation of DCMOP

DCMOP can be shown in Algorithm 1, including MOPF, data customization, and proxy mechanism. According to (3), the core of DCMOP is to accelerate the evaluation of subnetworks during the evolutionary process for network pruning, i.e., $M = \{M^{\text{pr}}(p_i), M^{\text{acc}}(p_i)\}$. Because the contradiction between the two indicators can be better balanced under the framework of MOPF. Therefore, DCMOP adopts the decomposition strategy and introduces the Tchebycheff method

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to solve this problem. The Tchebycheff can be expressed as follows:

$$\begin{cases} \min g^{\text{te}}(\mathbf{p} \mid \lambda_j, z^*) = \max\{\lambda_j \mid M_j(\mathbf{p}) - z_j^* \mid\}, \\ s.t. \quad x \in \Omega; 1 \geq i \leq m \end{cases} \tag{16}$$

where $z^* = \max M_j(p_i)$, $\mathbf{p} \in \Omega_p$ represents the reference point of some columns. For $j = \text{pr}, \text{acc}$, $z_j^* \geq \min\{M_j(p_i), p_i \in \Omega\}$.

For each solution, $\lambda_j = \{\lambda_{\text{pr}}, \lambda_{\text{acc}}\}$, and then we can obtain the PF by leveraging the value of λ iteratively. When evaluating each subnetwork with different pruning rates, the amount of data is customized according to the different parameters of the subnetworks to speed up the training of the subnetworks. Then, individuals along the PF can be used to provide potential subnetworks for the remote sensing scene classification. The advantages of the proposed method not only save time but also reduce the cost of relative engineering.

IV. Experiments

In order to verify the effectiveness of the proposed method, we test the four datasets of remote sensing image scene classification on two models (VGG-16 [24] and Resnet-50 [14]). This section is divided into three parts. The first part is the relevant description of the dataset and experimental details. The second part is the analysis experiments for VGG-16 [24], which includes the baseline results, the training results after pruning with the proposed method, and the result of ablation experiments. The third part is the analysis experiments for ResNet-50 [24], which includes the baseline results, the training results after pruning with the proposed method, and the result of ablation experiments. The fourth part is the analysis of the comparative experiment, which proves the effectiveness of our method by comparing it with other traditional pruning methods and NAS-based methods.

A. Datasets and Settings

In this section, we introduce the details of the four remote sensing scene classification datasets which as shown in Fig. 4, and explain the relevant details of experimental settings. In this section, the introduction of the four datasets and the experimental settings of the relevant details have been explained as follows.

1) Datasets: We mainly introduce the indicators of the dataset: 1) the input size of the dataset (the number of channels), which affects the amount of dataset required by the model; 2) the total size of the dataset, which is closely related to the size of model parameters; and 3) the total category of the dataset which is closely related to the accuracy of the classification results.

1) AID [10] contains 30 categories of scene images. There are about 220–420 images in each category, a total of 10 000 images in total, and the pixel size of each image is about 600×600. The dataset was released by Huazhong University of Science and Technology and Wuhan University in 2017.

2) NWPU-RESISC45 [11] is an available benchmark for remote sensing image scene classification created by the Northwest University of Technology. The dataset contains a total of 31 500 images with a pixel size of 256×256, covering

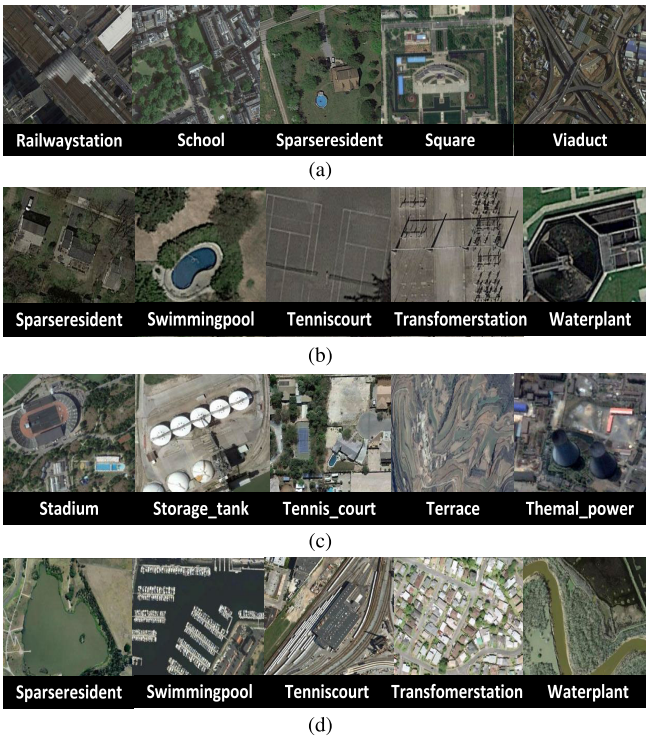


Fig. 4. (a)–(d) Lists some categories of AID, PatternNet, NWPU-RESISC45, and WHU-RS19 datasets, respectively.

45 scene categories, such as aircraft, airports, dense residential, desert, forest, and highway, of which 700 images are in each category.

3) PatternNet [12] is a large-scale high-resolution remote sensing dataset collected for remote sensing image retrieval. There are 38 categories in total, and each category has 800 pieces with a size of 256×256 pixel image. The images in PatternNet are collected from Google Earth images or through the Google Map API of some American cities.

4) WHU-RS19 [13] is a scene recognition dataset released by Wuhan University in 2011. The image of this dataset comes from the remote sensing image of China obtained from Google satellite images. The dataset has 19 scene categories, 50 pictures in each category, a total of 1005, and the size of each picture is 600×600 pixels.

2) Settings: All datasets are divided according to the ratio of 8:2, with 80% of the images as the training sets and 20% as the testing sets. In DCPM, the epoch required for the proxy model training is set to 20. The epoch required for the full training model is set to 150, which is selected from **EP**.

B. Analysis Experiments for VGG-16

The purpose of DCMOP is to balance the accuracy and pruning rate of the model, which needs to maintain higher accuracy possibly as well as reduce model size maximally. In this section, there are providing a detailed explanation of the result analysis on VGG-16, and then the optimal results are presented in bold to emphasize the comparison results more intuitively.

In the beginning, we conducted 150 generations of fully trained on the VGG-16 [24] original model and compared this as the baseline of the VGG-16 [24] model with

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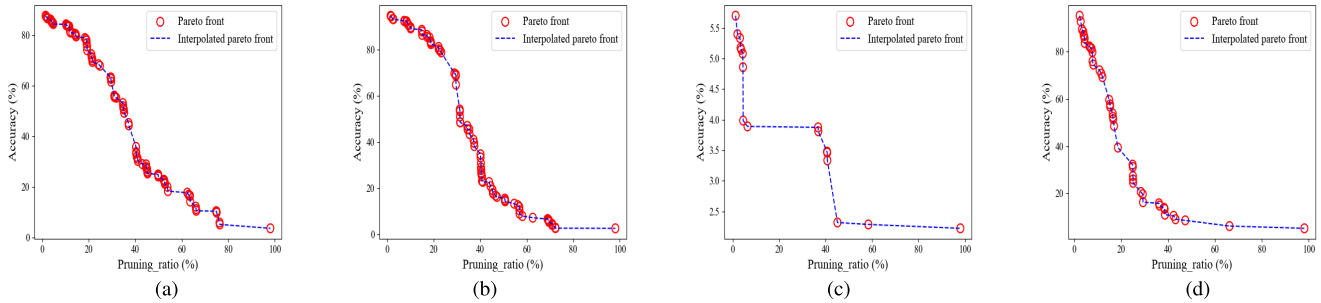


Fig. 5. EP of the proxy subnetwork generated by DCMOP on VGG-16. (a)–(d) Performance on AID, PatternNet, NWPU-RESISC45, and WHU-RS19, respectively.

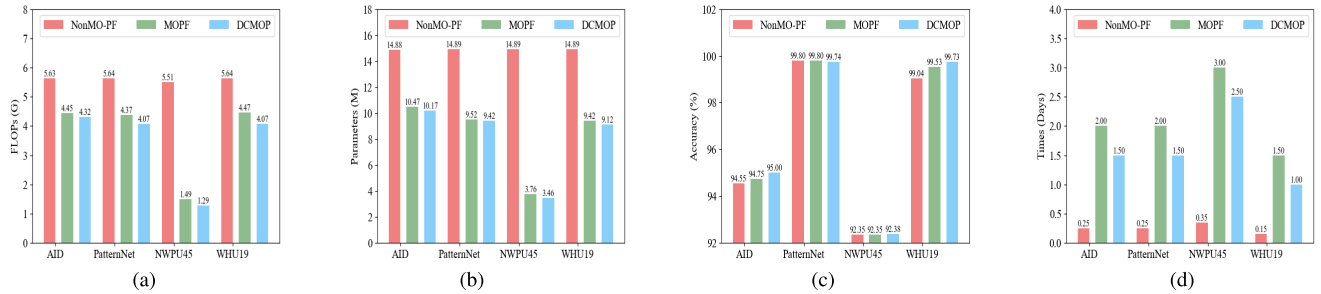


Fig. 6. Ablation experiment of DCMOP on VGG-16, including "NonMO-PF," "MOPF," and "DCMOP." (a)–(d) Results of FLOPs, parameters, accuracy, and times consumption of these three methods on four remote sensing datasets, respectively.

TABLE I

BASELINE OF VGG-16 AND RESNET-50 ON DIFFERENT DATASETS WHEN THE MODELS ARE TRAINED WITH 150 EPOCHS (FULLY TRAINING)

Models	Datasets	Param (M)	FLOPs (G)	Acc-150 (%)
VGG-16 [24]	AID [10]	17.25	7.71	93.25
	NWPU-RESISC45 [11]	17.25	7.71	92.65
	PatternNet [12]	17.25	7.71	97.78
	WHU-RS19 [13]	17.24	7.71	98.08
ResNet-50 [14]	AID [10]	25.78	2.06	94.50
	NWPU-RESISC45 [11]	25.78	2.06	91.84
	PatternNet [12]	25.78	2.06	98.12
	WHU-RS19 [13]	25.77	2.06	98.08

the fully-trained pruning subnetworks model generated by DCMOP. The results of the Baseline are shown in Table I.

In DCMOP, we constructed DCPM to accelerate the evaluation speed of subnetworks during the pruning process. Inspired by previous research [16], [47], [48], we find that incomplete training can leverage the shared structure or performance predictions among candidate architectures to rank the relative performance of candidate architectures to reflect their quality. Furthermore, incomplete training can avoid resource consumption while reflecting the quality of subnetwork structures compared to fully trained. Due to the positive correlation between incomplete and complete training of the network, incomplete training of subnetworks with different pruning rates generated in DCMOP can reflect the quality of fully trained subnetworks. Based on this conclusion, we regard the incompletely trained subnetwork as the proxy model.

Therefore, we regard the 20-generation incompletely trained subnetworks as the proxy model in DCMOP, and the **EP** of the proxy model generated after the evolution process is shown in Fig. 5, while not only the proxy model's responsiveness is optimal, but also can effectively reduce the cost in the evolution process.

Due to the individual from the Pareto frontier solutions generated by DCMOP having the same potential, we fully train them for 150 generations, and then select five individuals who perform well to present as results in Table II. We can observe that the pruning rate representation of VGG-16 [24] models varies on different datasets, which fully proves the robustness of DCMOP for remote sensing image scene classification on VGG-16 [24] models. For AID [10], the model with a pruning rate of 10.3% is the best, with an accuracy of 95.00%. This model not only outperforms the baseline model with an accuracy of 93.25% terms of accuracy but also reduces FLOPS and parameters by 44.10% and 41.40%, respectively. For PatternNet [12], the accuracy of the model with a pruning rate of 4.05% is 99.74%. This model not only outperforms the baseline model with an accuracy of 97.78% but also reduces floats and parameters by 47.86% and 47.62%, respectively. For NWPU-RESISC45 [11], the model with a pruning rate of 3.48% is the best, with an accuracy of 92.38%. The accuracy of the model remains basically unchanged, but the FLOPs and parameters are reduced by 83.27% and 79.94% compared to the original model, respectively, which effectively achieves a balance between model accuracy and pruning rate. For WHU-RS19 [13], a model with a pruning rate of 6.25% is the best, with an accuracy of 99.51%. This model not only has higher accuracy than the original model with an accuracy of 98.08% but also reduces floats and parameters by 55.90% and 52.90%, respectively. Overall, DCMOP can find the model that best

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TABLE II

RESULT OF THE FULLY-TRAINED VGG-16 ON DIFFERENT DATASETS, WHICH SELECTS FIVE HIGH-QUALITY PROXY MODEL SUBNETWORKS IN EP THROUGH THE PROXY MECHANISM, RESPECTIVELY

Models (Epochs)	Datasets	Pruning-Ratio (%)	FLOPs (G)	Params (M)	Accu(proxy) (%)	Accu(real) (%)	Accu-O (%)
VGG-16(150)	AID [10]	5.15	4.54	10.49	76.35	94.05	93.25
		5.90	4.54	10.45	76.23	93.95	
		3.60	7.39	16.65	98.11	94.60	
		10.30	**4.31**	**10.15**	**74.93**	**95.30**	
		9.80	4.32	10.17	75.00	95.00	
	NWPU-RESISC45 [11]	2.22	7.43	16.62	98.04	93.59	92.65
		2.32	1.55	4.16	45.08	92.40	
		3.33	1.30	3.49	40.73	92.21	
		3.46	1.30	3.48	40.67	91.92	
		3.48	**1.29**	**3.46**	**40.60**	**92.38**	
	PatternNet [12]	2.63	7.43	16.65	98.13	99.78	97.78
		2.70	4.09	9.53	72.37	99.76	
		2.93	4.24	9.52	72.35	99.76	
		4.05	**4.07**	**9.12**	**99.73**	**99.74**	
		4.56	4.02	9.13	70.95	99.70	
	WHU-RS19 [13]	5.29	7.44	16.64	98.13	98.55	98.08
		6.25	**3.40**	**8.12**	**66.19**	**99.51**	
		8.65	1.82	4.47	47.35	96.15	
		9.13	1.51	3.79	43.21	95.19	
		10.58	1.39	3.71	42.40	88.94	

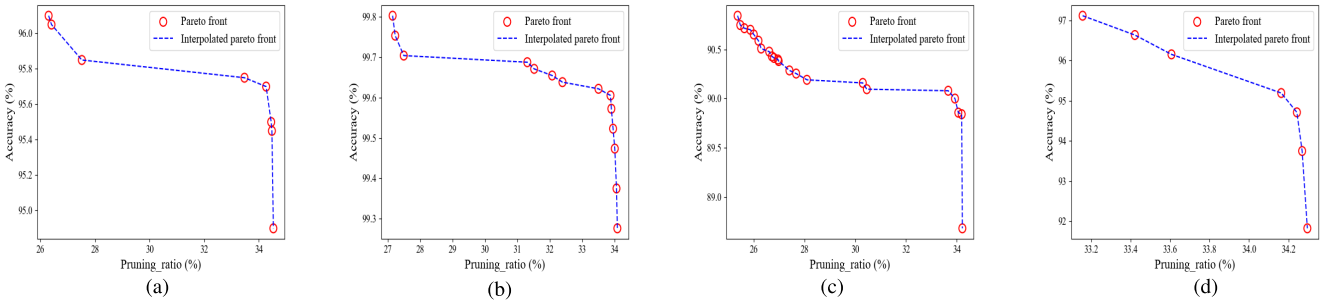


Fig. 7. EP of the proxy subnetwork generated by DCMOP on ResNet-50. (a)–(d) Performance on AID, PatternNet, NWPU-RESISC45, and WHU-RS19, respectively.

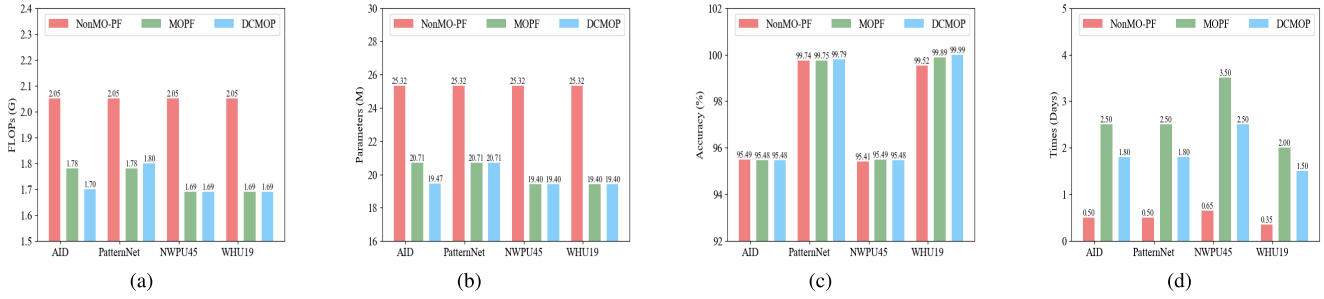


Fig. 8. Ablation experiment of DCMOP on ResNet-50, including "NonMO-PF," "MOPF," and "DCMOP." (a)–(d) Results of FLOPs, parameters, accuracy, and times consumption of these three methods on four remote sensing datasets, respectively.

balances accuracy and pruning rate on the VGG-16 [24] model for different datasets, which demonstrates its effectiveness and robustness.

Ablation Experiment: We conduct "NonMO-PF," "MOPF," and "DCMOP" on VGG-16 [24] to illustrate the efficacy of the data customization mechanism. "NonMO-PF" prunes by manually adjusting to set a suitable pruning truncation through multiple experiments. MOPF is a pruning framework with multiobjective optimization. DCMOP is a multiobjective pruning framework that utilizes data customization mechanism

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TABLE III
RESULT OF THE FULLY-TRAINED RESNET-50 ON DIFFERENT DATASETS, WHICH SELECTS FIVE HIGH-QUALITY PROXY MODEL SUBNETWORKS IN EP THROUGH THE PROXY MECHANISM, RESPECTIVELY

Models (Epochs)	Datasets	Pruning-Ratio (%)	FLOPs (G)	Params (M)	Accu(proxy) (%)	Accu(real) (%)	Accu-O (%)
ResNet-50(150)	AID [10]	27.52	1.77	20.49	95.85	96.20	94.50
		34.27	1.69	19.33	95.70	96.45	
		26.30	1.80	20.72	96.10	96.70	
		33.47	**1.70**	**19.47**	**95.75**	**96.40**	
		26.40	1.80	20.72	26.40	96.05	
	NWPU-RESISC45 [11]	25.64	1.80	20.76	90.71	93.59	91.84
		25.48	1.81	20.76	90.75	95.16	
		26.01	1.8	20.75	90.65	95.51	
		25.37	1.81	20.76	90.84	95.32	
		25.87	**1.80**	**20.71**	**90.70**	**95.48**	
	PatternNet [12]	31.30	1.685	19.4	99.69	99.81	98.12
		31.51	**1.69**	**19.40**	**99.67**	**99.79**	
		27.22	1.77	20.59	99.75	99.80	
		27.15	1.79	20.57	99.80	99.85	
		27.49	1.77	20.48	99.70	99.82	
	WHU-RS19 [13]	34.16	1.70	19.47	95.19	99.49	98.08
		33.16	1.70	19.63	97.11	99.52	
		33.42	1.70	19.44	96.63	99.52	
		34.24	**1.69**	**19.40**	**94.71**	**99.99**	
		33.61	1.69	19.48	96.15	99.51	

acceleration. We compared the four indicators of these three methods: FLOPs, parameters, accuracy, and time consumption.

The comparison results are shown in Fig. 6. Compared with NonMO-PF, MOPF can better balance the pruning rate of the model under the condition of guaranteeing the accuracy of the model. As shown in Fig. 6(a) and (b), there is a significant reduction in FLOPs and Parameters of the model using MOPF. Thus, it proves that the introduction of multi-objective optimization into the pruning framework can well balance the contradiction between the accuracy of the model and the pruning rate. As shown in Fig. 6(a) and (b), DCMOP significantly reduces FLOPs and parameters compared to the other two methods, which can improve the efficiency of the model. Especially, DCMOP has different effects on different datasets and is more targeted. As shown in Fig. 6(c), compared with the other two methods, DCMOP has good precision in different datasets. As shown in Fig. 6(d), DCMOP has lower time consumption compared to MOPF, and the strategy of data customization can effectively accelerate the iterative evaluation speed of MOPF, thereby reducing costs. Therefore, the effectiveness of DCMOP has been demonstrated through ablation experiments.

C. Analysis Experiments for ResNet-50

The ResNet-50 [14] is a stack of convolutional layers, which includes 49 convolutional layers and one fully connected layer. Due to the fact that ResNet-50 [14] has more convolutional layers than the VGG-16 [24], it can demonstrate better accuracy in scene classification of remote sensing images compared to the VGG-16 [24]. Nevertheless, due to the deeper depth of the ResNet-50 [14] compared with the VGG-16 [24], the parameter quantity of the ResNet-50 [14] is relatively large, which can affect the efficiency of remote sensing image scene classification and excessively consume computational resources. Experiments on ResNet-50 [14] can show that DCMOP can effectively solve the above problems. The optimal results are presented in bold to emphasize the comparison results more intuitively.

The same operation on ResNet-50 [14] as VGG-16 [24] is performed. In the beginning,

The original model on ResNet-50 [14] with fully trained for 150 generations is regarded as the baseline of the model, which is shown in Table I. For DCMOP, an incompletely trained initialized pruning subnetwork is used as a proxy model on the ResNet-50 [14] to reflect the quality of the pruning subnetwork during complete training. Among them, the PF performance of the proxy model on the AID [10], PatternNet [12], NWPU-RESISC45 [11], and WHU-RS19 [13] is shown in Fig. 7.

In the generated PF, each individual has the same potential, and then the subnetworks have been fully trained. However, subnetworks that ensure accuracy and have a high pruning rate are superior. Therefore, five well-performing subnetworks are presented as results in Table III. We can observe that the pruning rate of the ResNet-50 [14] model exhibits different performance on different datasets, which fully proves the robustness of DCMOP in remote sensing scene classification on the ResNet-50 [14] model.

In addition, ResNet-50 [14] has higher accuracy than VGG-16 [24]. Especially in the process of pruning subnetworks, ResNet-50 [14] performs more prominently. For AID [10], the model with a pruning rate of 33.47% has the best effect,

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TABLE IV
COMPARISONS OF DIFFERENT COMPRESSION TECHNIQUES WITH OUR METHOD (DCMOP). THE "GDAYS" MEANS THE TIME COSTS OF THE CORRESPONDING METHODS WITH ONE 2080TI

Dataset		Method	Accuracy	FLOPs	Parameters	Searching Cost (GDays)
AID [10]	NAS-Based	Automatically Architecture [49]	91.12%	-	3.80M	-
		SLGE-based [6]	96.10%	0.56G	5.10M	9.60
	Pruning-Based	Taylor Pruning [18]	96.15%	1.66G	18.13M	0.13
		Network Slimming [19]	96.15%	1.81G	18.02M	0.15
		SEP [50]	96.10%	1.56G	17.93M	0.14
		EFPF ($\alpha = 0.0020$) [51]	96.90%	1.87G	17.98M	0.13
		DCMOP	**97.00%**	**1.79G**	**15.49M**	**1.80**
NWPU-RESISC45 [11]	NAS-Based	SceneNet [52]	95.22%	9.47G	1.02M	10.00
		SLGE-based [6]	96.56%	0.56G	5.10M	9.60
		RSNet [53]	92.66%	1.28G	1.40M	2.00
	Pruning-Based	Taylor Pruning [18]	94.80%	1.59G	18.29M	0.13
		Network Slimming [19]	94.53%	1.93G	17.66M	0.15
		SEP [50]	94.65%	1.50G	18.08M	0.14
		EFPF ($\alpha = 0.0020$) [51]	95.02%	1.87G	17.99M	0.13
		DCMOP	**95.48%**	**1.785G**	**20.71M**	**1.80**
PatternNet [12]	Pruning-Based	Taylor Pruning [18]	99.80%	1.47G	18.33M	0.12
		Network Slimming [19]	99.85%	1.86G	17.42M	0.13
		SEP [50]	99.70%	1.43G	17.92M	0.14
		EFPF ($\alpha = 0.0020$) [51]	99.47%	1.87G	17.99M	0.12
		DCMOP	**99.79%**	**1.79G**	**19.40M**	**2.50**
WHU-RS19 [13]	Pruning-Based	Taylor Pruning [18]	98.73%	1.71G	17.78M	0.05
		Network Slimming [19]	98.91%	1.88G	16.97M	0.05
		SEP [50]	98.63%	1.62G	17.58M	0.14
		EFPF ($\alpha = 0.0020$) [51]	99.90%	1.87G	17.99M	0.05
		DCMOP	**99.99%**	**1.69G**	**19.40M**	**1.50**

with an accuracy of 96.40%. This model not only improves the accuracy of the baseline model by 1.9% but also reduces FLOPs and parameters by 17.48% and 24.48%, respectively. For NWPU-RESISC45 [11], the pruning effect is best with a pruning rate of 25.87% and an accuracy of 95.48%. This model not only improves the accuracy of the baseline model by 2.83% but also reduces FLOPs and parameters by 13.35% and 19.67%, respectively. For PatternNet [12], the pruning effect is best with a pruning rate of 31.51% and an accuracy of 99.79%. This model not only improves the accuracy of the baseline model by 1.67% but also reduces FLOPs and parameters by 18.20% and 24.75%, respectively. For WHU-RS19 [13], the pruning effect is best with a pruning rate of 34.24% and an accuracy of 99.99%. This model not only improves the accuracy of the baseline model by 1.91% but also reduces FLOPs and parameters by 18.20% and 24.75%, respectively.

Ablation Experiment: We conduct "NonMO-PF," "MOPF," and "DCMOP" on ResNet-50 [14] to illustrate the efficacy of the data customization mechanism. "NonMO-PF" means the pruning framework without applying multiobjective optimization techniques, and the "MOPF" implies only the conventional multiobjective optimization algorithm is applied to the pruning task of CNNs. Same the ablation experiment of VGG-16 [24], we also compared the four indicators of these three methods on ResNet-50 [14]: FLOPs, parameters, accuracy, and time consumption.

As shown in Fig. 8(a) and (b), there is a significant reduction in FLOPs and Parameters of the model using MOPF on all datasets. Thus, it proves that the introduction of multiobjective optimization into the pruning framework can well balance the contradiction between the accuracy of the model and the pruning rate. As shown in Fig. 8(a) and (b), DCMOP can reduce the FLOPs and parameters of the model more effectively than the other two methods, and has a better structural effect than VGG-16 [24]; as shown in Fig. 8(c), the accuracy of DCMOP has been significantly improved. As shown in Fig. 8(d), although the time of ResNet-50 [14] has been extended, the cost is not high.

D. Comparisons With NAS and Other Pruning Methods

In this section, we compared DCMOP with other pruning methods and NAS-based methods on each dataset. Due to the superior performance of each dataset on ResNet-50 [14] compared to the results on VGG-16 [24], we compared the

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performance of DCMOP with the results of other methods on ResNet-50 [14]. The comparison results are shown in Table IV. To emphasize the comparison results more intuitively, DCMOP is presented in bold.

For AID [10], the two NAS-based approaches, "automatically architecture" [49] and "SLGE-based" [6], are effective in reducing flops and parameters. However, the search time consumption of these NAS-based methods is very costly. Compared with these methods, DCMOP not only improves the accuracy by about 5% but also reduces the search cost-effectively. What's more, the traditional pruning methods of "Taylor pruning" [18], "SEP" [50], "EFPF" [51], and "network slimming" [19] are less expensive, but it is difficult to reduce the parameters of the model. DCMOP can ensure the expressiveness of the model while effectively reducing FLOPs and Parameters. Especially, it is particularly evident in the Parameter indicator of the model, which drops by 3M compared to traditional pruning methods; for NWPU-RESISC45 [11], compared to "SceneNet" [32], "SLGE-based" [6], and "RSNet" [53], DCMOP not only has comparable model performance but also reduces the search cost by up to 5 times. This means that DCMOP not only balances the accuracy and pruning power of the model but also effectively improves the evaluation efficiency and thus reduces the search cost. DCMOP is able to better balance model accuracy and pruning rate than traditional pruning methods ("Taylor pruning" [18], "SEP" [50], "EFPF" [51], and "network slimming" [19]); for PatternNet [12], although "Taylor pruning" [18], "SEP" [50], "EFPF" [51], and "network slimming" [19] perform a good precision, it is obvious that these two methods have unbalanced roles in FLOPs and Parameter. DCMOP can solve this problem well; for WHU-RS19 [13], the accuracy of the model in DCMOP is better than that of "Taylor pruning" [18], "SEP" [50], "EFPF" [51], and "network slimming" [19]. In addition, the Flops of the model in DCMOP are 0.35M and 0.195M lower than that of the model in "Taylor pruning" [18] and "network slimming" [19], respectively. This means that DCMOP not only ensures the expressiveness of the model but also improves the efficiency of the model.

Through the above experimental results, it can be concluded that although NAS-based methods have good results in reducing the FLOPs and parameters of the model, however, these methods are very time-consuming leading to excessive search costs and thus waste of computational resources. Since traditional pruning methods are weight-sensitive, a pruning parameter is usually designed manually to prune the model. Therefore, traditional pruning methods often have difficulty in balancing the accuracy of the model and the pruning rate, and thus cannot effectively reduce the FLOPs and parameters of the model.

Impressively, DCMOP not only has lower search costs than NAS-based methods but also has higher accuracy than NAS-based methods. In addition, compared to traditional pruning methods such as "Taylor pruning" [18], "SEP" [50], "EFPF" [51], and "network slimming" [19], DCMOP not only has better accuracy but also reduces Flops and parameters, effectively improving the efficiency of the model. In particular, DCMOP is able to have different pruning effects on different remote sensing image datasets and is more robust to remote sensing scene classification.

V. Conclusion

In order to balance the accuracy and the pruning rate of the model, in this article, we propose a DCMOP framework to prune models. The multiobjective optimization pruning model can be used to find the pruning model with the maximum pruning rate that can ensure accuracy through iterative evolution in the generated subnetworks with different pruning rates. This method can effectively avoid the drawbacks of manual pruning rate parameters, making the pruning framework more robust. However, the evaluation process of subnets by MOPF during pruning can waste time, which may lead to the limitation of satellite resource platforms. Therefore, we designed DCPM to accelerate the evaluation speed and reduce time costs. Because different pruning subnetworks have different parameter amounts, the data required for the subnetwork is customized according to the size of the data amount through data self-adaptation, thereby accelerating the speed of subnetwork training. At the same time, we constructed a proxy model reflection mechanism to accelerate the evolution process and reduce time costs.

We have applied VGG-16 [24] and Resnet-50 [14] to the dataset of remote sensing scene classification for AID [10], NWPU-RESISC45 [11], PatternNet [12], and WHU-RS19 [13], respectively, and then achieved good results, thereby proving that the method proposed in this article is effective and robust. According to the experimental results, DCMOP can not only effectively prune the model to reduce the number of parameters in the model, but also effectively reduce costs. This means that our method can effectively avoid manual adjustment of pruning parameters in traditional pruning methods, and is more robust. In addition, compared to NAS-based methods, DCPM in DCMOP effectively accelerates the evaluation speed of the model and finds the optimal subnetwork, which can effectively reduce time costs.

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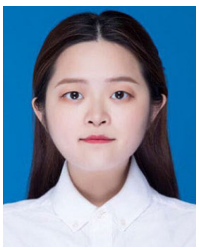
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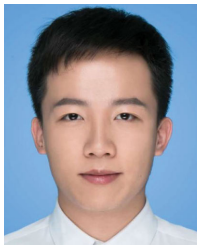
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